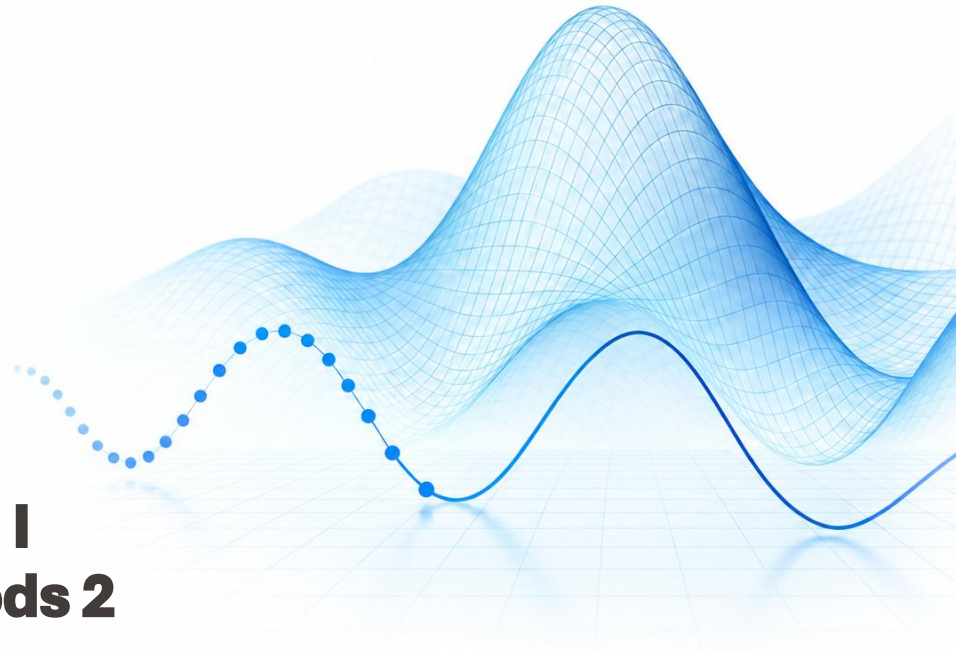


IE2106 – Engineering Math I
Week 8: Numerical Methods 2

Mohammad NIKOO
Nanyang Assistant Professor
Singapore NRF Fellow
EEE, NTU

What We Learnt Last Week

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Polynomial Interpolation

Newton's Divided Difference Method

We want to approximate a function $f(x)$ using an nth-degree polynomial $p_n(x) = a_n x^n + \cdots + a_1 x + a_0$

We select $n+1$ points from $f(x)$: $(x_0, f_0),\ \ldots,\ (x_n, f_n)$

We use the Newton's divided difference technique to form the polynomial. For example, for $n = 2$:

$$p_2(x) = f_0 + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

Upper bound of error at any x

$$|f(x) - p_n(x)| \leq \frac{1}{(n+1)!}\left|\prod_{k=0}^{n}(x - x_k)\right| \max_{\xi}\left|f^{(n+1)}(\xi)\right|$$

$$\min\{x, x_0, x_1, \ldots, x_n\} \leq \xi \leq \max\{x, x_0, x_1, \ldots, x_n\}$$

Numerical Solution of Equations

Fixed-Point Iteration with n Correct Decimals

We find a sequence converging to the root of $f(x) = 0$. Here is what we need to do:

1) Find an interval $J = [a, b]$ that includes a root. Criterion for continuous function f: $f(a) \cdot f(b) < 0$

2) Re-write the equation in this form: $x = g(x)$

3) Check the convergence: If the iteration remains in J, then the condition is $\max_{a \leq x \leq b} |g'(x)| < 1$

4) Calculate $m = \min_{a \leq x \leq b} |f'(x)|$

5) Form the sequence: $x_{n+1} = g(x_n)$ and calculate both x_n and $\frac{|f(x_n)|}{m}$. Stop when $\frac{|f(x_n)|}{m} < \frac{1}{2} \times 10^{-K}$

Today's Objectives

Our main objective today is to numerically solve Ordinary Differential Equations (ODEs), for example:

$$y' + 2y - 3 = 0, \qquad 2y' - y^2 + x = 1, \qquad y'' + \frac{1}{x^2+1}y' - y = x$$

- These equations may or may not have analytical (exact) solutions.
- We aim to approximately solve these equations using numerical methods.
- It is true that our method is not exact, but it can be generally applied to ODEs. So it is quite useful!

We will learn three numerical methods

- Euler Method
- Improved Euler Method
- Fourth-order Runge–Kutta Method

PART I

NUMERICAL METHODS TO SOLVE FIRST-ORDER ODE

Why do we need numerical methods?

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There are ODEs that we can't solve

Analytical approaches work for specific types of ODEs!

Example

Abel equation of the first kind

$$\frac{\mathrm{d}y}{\mathrm{d}t} = y^3(t) + t$$

Nonlinear spring-mass system

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \delta\frac{\mathrm{d}x}{\mathrm{d}t} + \alpha x + \beta x^3 = \gamma\cos(\omega t)$$

Sometimes the solution is in integral form

We can solve the ODE analytically, but the analytical solution is very complex!

Example

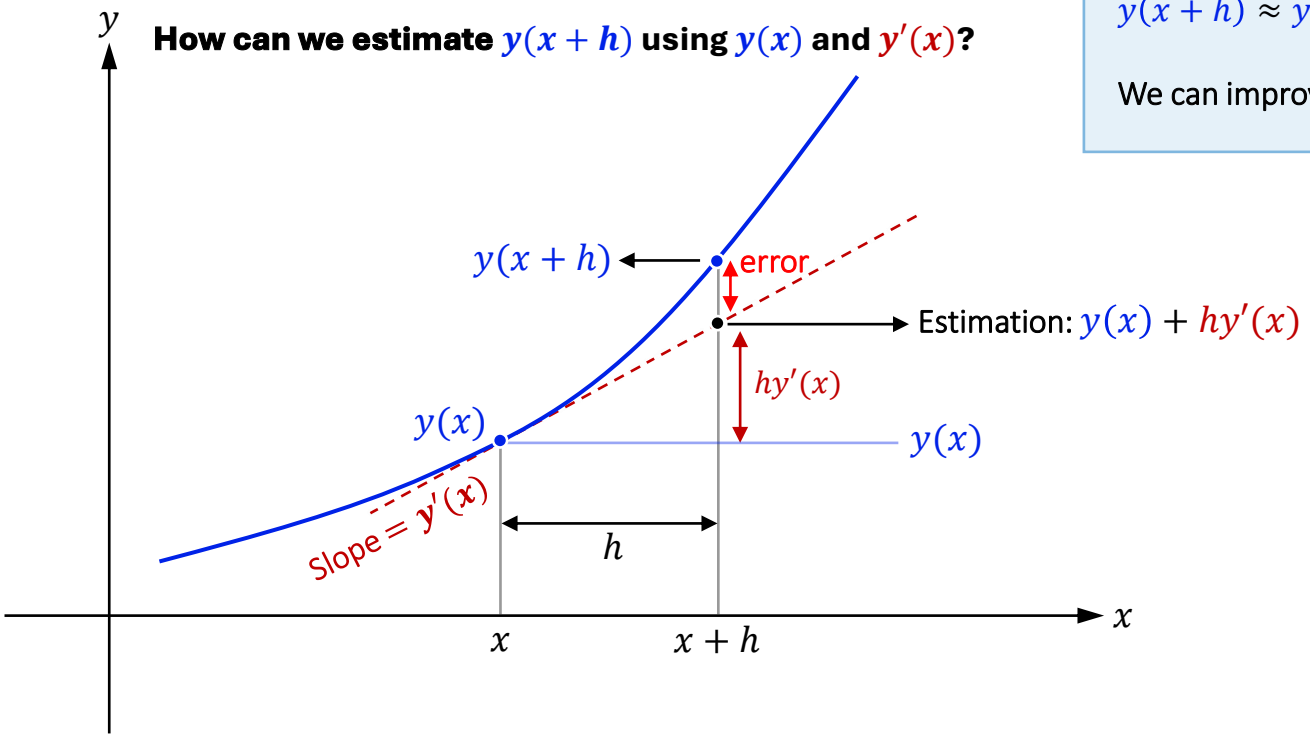
$$\frac{\mathrm{d}y}{\mathrm{d}t} = \sin(t) + \cos(t)\,y(t) + e^t y^2(t)$$

The solution can be written as

$$y(t) = -e^{-t}\frac{C_1u_1'(t) + C_2u_2'(t)}{C_1u_1(t) + C_2u_2(t)}$$

$$u_2(t) = u_1(t)\int^t \frac{W_0 e^{s+\sin(s)}}{u_1^2(s)}\,\mathrm{d}s$$

Core Idea: Estimation Using Derivative



$$y(x+h) \approx y(x) + hy'(x)$$

We can improve the accuracy by reducing h

Euler's Method

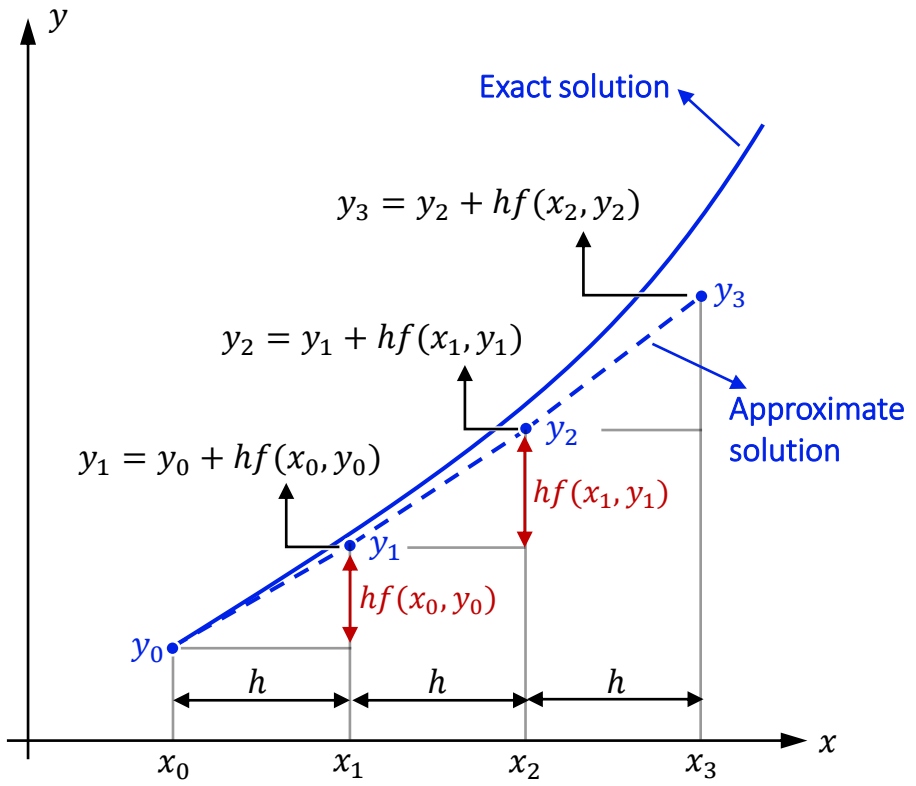
We consider a first-order ODE in a general form

$$y' = f(x, y), \qquad y(x_0) = y_0$$

f is a given function and y_0 is a given initial condition.

Euler's method considers a step size of h. It approximately computes $y_n \approx y(x_n)$ for $x_n = x_0 + nh$.

x_n	y_n (approximate solution at $x = x_n$)
x_0 (known)	y_0 (known)
$x_1 = x_0 + h$	$y_1 = y_0 + y_0' h = y_0 + f(x_0, y_0)h$
$x_2 = x_0 + 2h$	$y_2 = y_1 + y_1' h = y_1 + f(x_1, y_1)h$
...	...
$x_{n+1} = x_0 + (n+1)h$	$y_{n+1} = y_n + y_n' h = y_n + f(x_n, y_n)h$



Euler's Method: Example

Solve $y' = e^{y+\sin(t)} - 1$ with initial condition $y(0) = 0$ using Euler's method with step size of $h = 0.1$.

We want to find $y(t_n)$ where $t_n = nh \ \ (n = 0, 1, 2, 3, \ldots)$

$t_0 = 0 \rightarrow \boxed{y(0) = 0}$ (This is our initial condition!)

$t_1 = 0.1 \rightarrow y(0.1) \approx y(0) + y'(0)h = 0 + \left(e^{y(0)+\sin(0)} - 1\right) \times 0.1 = 0 + \left(e^{0+0} - 1\right) \times 0.1 = 0$

$y(0) = 0$ (Initial condition)

$y'(0)$ can be calculated using the ODE

$\boxed{y(0.1) \approx 0}$

$t_2 = 0.2 \rightarrow y(0.2) \approx y(0.1) + y'(0.1)h \approx y(0.1) + \left(e^{y(0.1)+\sin(0.1)} - 1\right)h \approx 0 + \left(e^{0+\sin(0.1)} - 1\right) \times 0.1 \approx 0.010499$

$y'(0.1)$ can be calculated using the ODE

$\boxed{y(0.2) \approx 0.010499}$

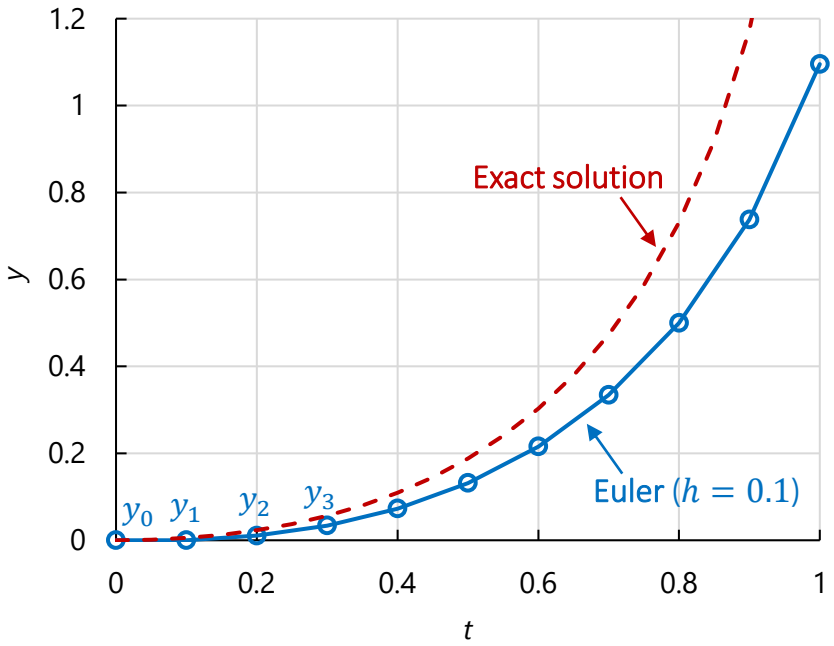
$t_3 = 0.3 \rightarrow y(0.3) \approx y(0.2) + y'(0.2)h \approx y(0.2) + \left(e^{y(0.2)+\sin(0.2)} - 1\right)h \approx 0.010499 + \left(e^{0.010499+\sin(0.2)} - 1\right) \times 0.1 \approx 0.033764$

$y'(0.2)$ can be calculated using the ODE

$\boxed{y(0.3) \approx 0.033764}$

Euler's Method: Example

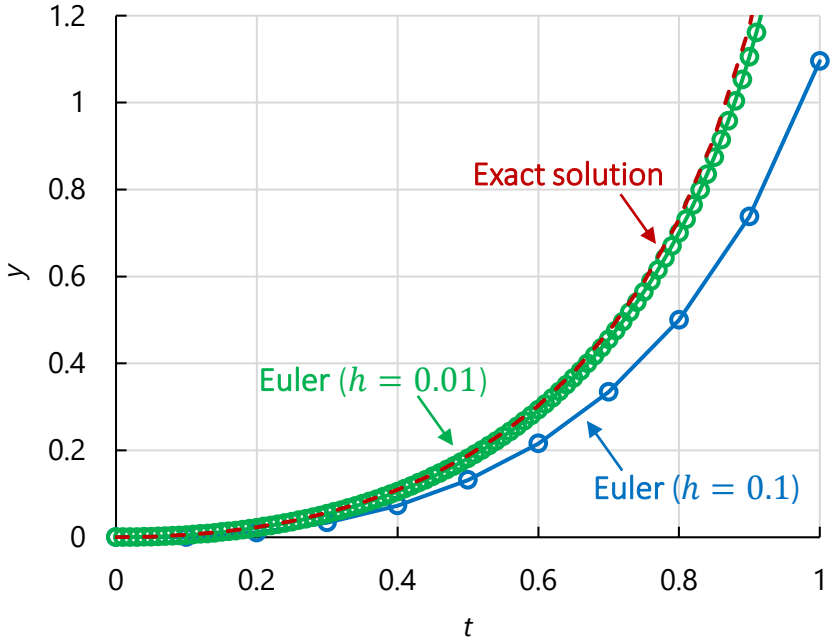
Solve $y' = e^{y+\sin(t)} - 1$ with initial condition $y(0) = 0$ using Euler's method with step size of $h = 0.1$.



t	Euler approximation y
0.0	0.000000
0.1	0.000000
0.2	0.010499
0.3	0.033764
0.4	0.072761
0.5	0.131514
0.6	0.215730
0.7	0.333959
0.8	0.499919
0.9	0.737716
1.0	1.095416

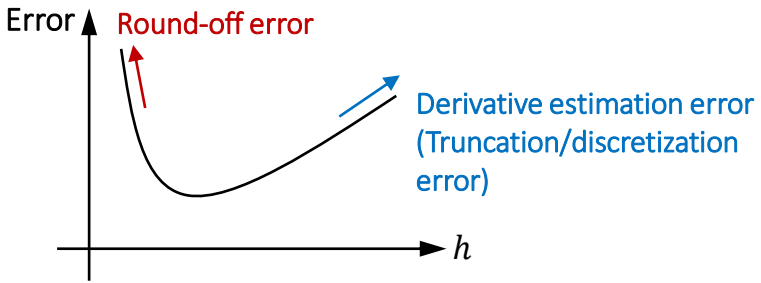
Euler's Method: Example

Solve $y' = e^{y+\sin(t)} - 1$ with initial condition $y(0) = 0$ using Euler's method with step size of $h = 0.1$.

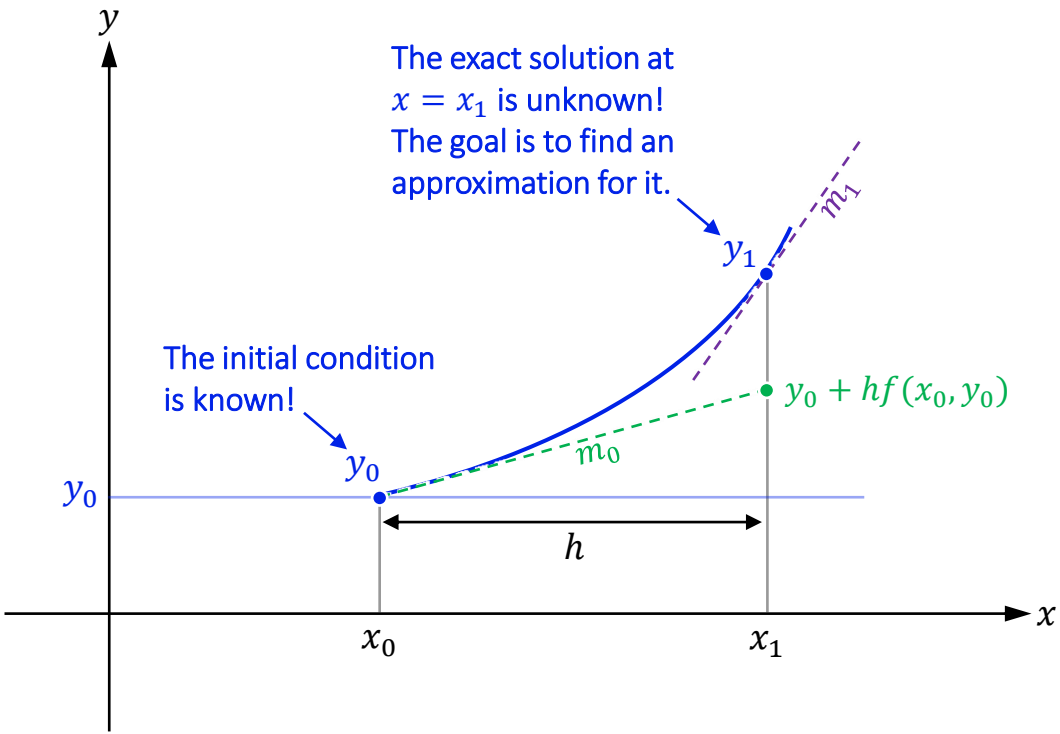


The error gets larger as t deviates more from t_0. Why?
Because we produce error in each calculation.
y_n carries the error of y_{n-1} and adds additional error

Is it better to keep making h smaller and smaller?
As more steps are taken, local errors can propagate and accumulate. Reducing h generally reduces the discretization error, but very small h requires many more calculations, so round-off error can eventually become significant..



Improved Euler's Method



Let's consider the following ODE and initial condition:

$$y' = f(x, y), \qquad y(x_0) = y_0$$

Slope at x_0

$$m_0 = f(x_0, y_0)$$

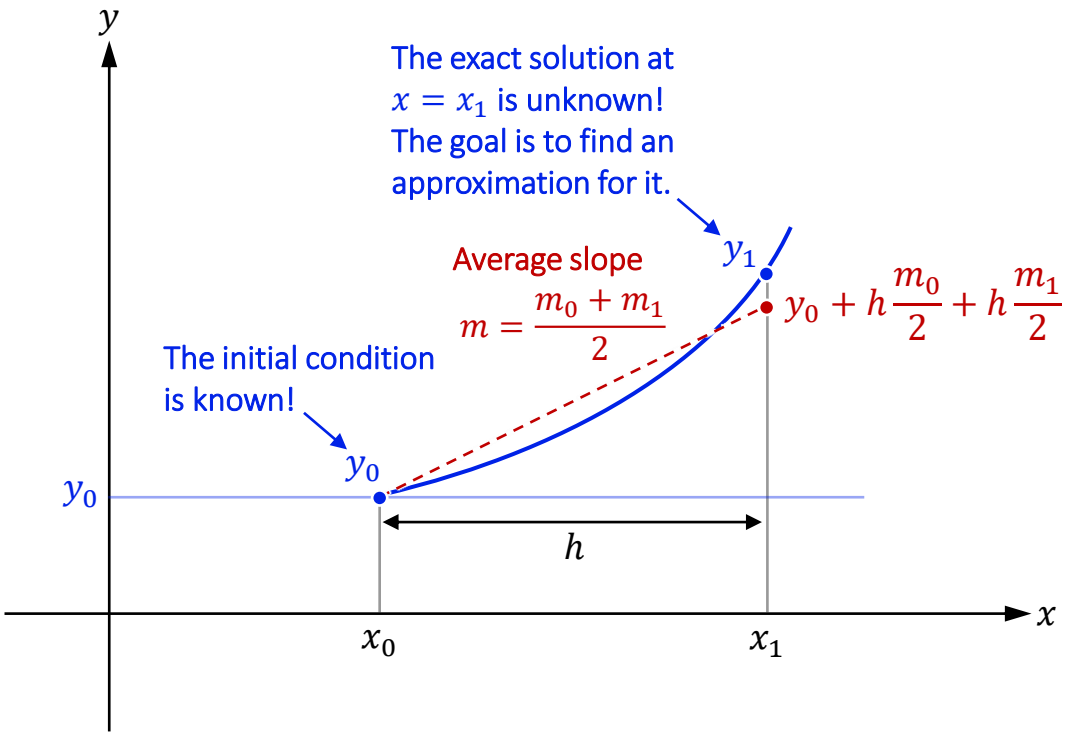
Approximation of slope at x_1

$$m_1 = f(x_1, y_1) \approx f(x_1, y_0 + hf(x_0, y_0))$$

We don't know y_1

We replace y_1 by its approximation

Improved Euler's Method



Let's consider the following ODE and initial condition:

$$y' = f(x, y), \qquad y(x_0) = y_0$$

Slope at x_0

$$m_0 = f(x_0, y_0)$$

Approximation of slope at x_1

$$m_1 = f(x_1, y_1) \approx f(x_1, y_0 + hf(x_0, y_0))$$

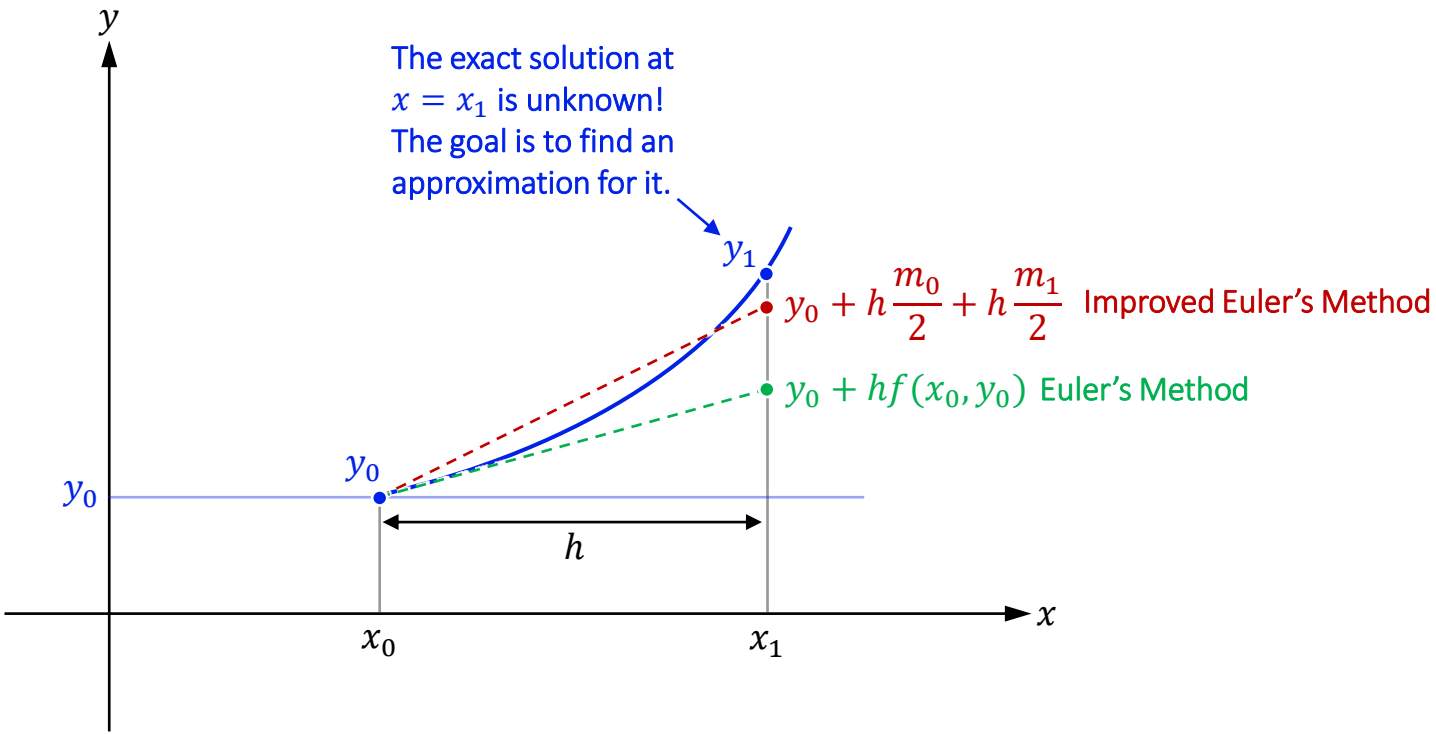
Average slope between x_0 and x_1

$$m = \frac{m_0 + m_1}{2}$$

The improved Euler's approximation at x_1 is as follows

$$\boldsymbol{y_1 \approx y_0 + h\frac{m_0}{2} + h\frac{m_1}{2}}$$

Euler's Method vs. Improved Euler's Method



Improved Euler's Method

For $y' = f(x, y)$, with $y(x_0) = y_0$, improved Euler's method computes approximations $y_n \approx y(x_n)$ at $x_n = x_0 + nh$.

x	a Increment using the slope at the starting point	b Increment using the slope at the predicted endpoint	y
x_0	—	—	y_0 (known!)
$x_1 = x_0 + h$	$a_0 = f(x_0, y_0)h$	$b_0 = f(x_1, y_0 + a_0)h$	$y_1 = y_0 + \frac{1}{2}(a_0 + b_0)$
$x_2 = x_0 + 2h$	$a_1 = f(x_1, y_1)h$	$b_1 = f(x_2, y_1 + a_1)h$	$y_2 = y_1 + \frac{1}{2}(a_1 + b_1)$
⋮	⋮	⋮	⋮
x_{n+1} $= x_0 + (n+1)h$	$a_n = f(x_n, y_n)h$	$b_n = f(x_{n+1}, y_n + a_n)h$	$y_{n+1} = y_n + \frac{1}{2}(a_n + b_n)$

Class Exercise

Use improved Euler's method with step size $h = 0.1$ to approximate $y(0.2)$, given

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x + y, \qquad y(0) = 1.$$

Class Exercise

Use improved Euler's method with step size $h = 0.1$ to approximate $y(0.2)$, given

$$\frac{dy}{dx} = x + y, \qquad y(0) = 1.$$

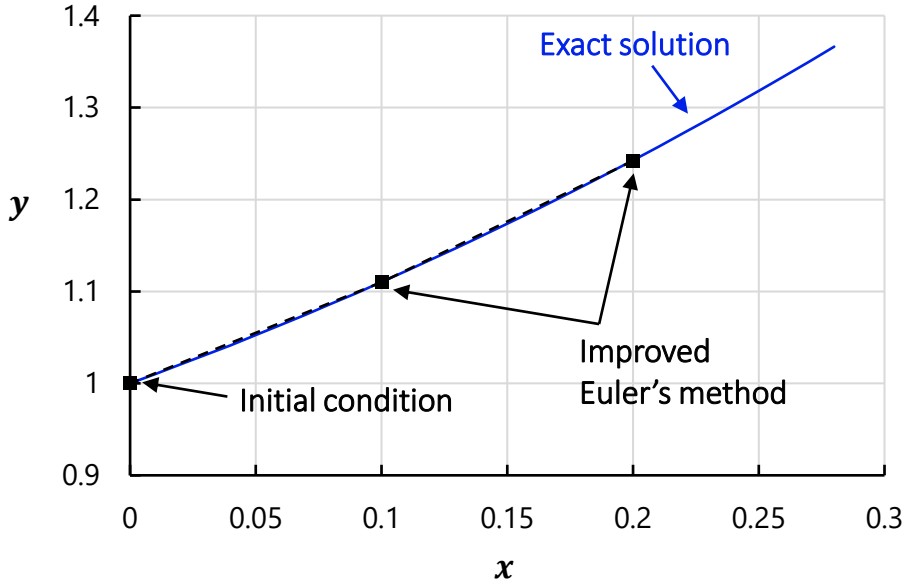
We have $f(x, y) = x + y$, $x_0 = 0$, and $y_0 = 1$. We can form the table:

x	a Increment using the slope at the starting point	b Increment using the slope at the predicted endpoint	y
$x_0 = 0$	—	—	$y_0 = 1$
$x_1 = 0.1$	$a_0 = f(x_0, y_0)h$ $= f(0,1) \cdot 0.1$ $= (0 + 1) \cdot 0.1$ $= 0.1$	$b_0 = f(x_1, y_0 + a_0) \cdot 0.1$ $= f(0.1,1 + 0.1) \cdot 0.1$ $= (0.1 + 1.1) \cdot 0.1$ $= 0.12$	$y_1 = y_0 + \frac{1}{2}(a_0 + b_0)$ $= 1 + \frac{1}{2}(0.1 + 0.12) = 1.11$
$x_2 = 0.2$	$a_1 = f(x_1, y_1)h$ $= f(0.1,1.11) \cdot 0.1$ $= (0.1 + 1.11) \cdot 0.1$ $= 0.121$	$b_1 = f(x_2, y_1 + a_1)h$ $= f(0.2,1.11 + 0.121) \cdot 0.1$ $= (0.2 + 1.231) \cdot 0.1$ $= 0.1431$	$y_2 = y_1 + \frac{1}{2}(a_1 + b_1)$ $= 1.11 + \frac{1}{2}(0.121 + 0.1431) = 1.24205$

Class Exercise

Use improved Euler's method with step size $h = 0.1$ to approximate $y(0.2)$, given

$$\frac{dy}{dx} = x + y, \qquad y(0) = 1.$$



x	y (Improved Euler's method)
$x_0 = 0$	$y_0 = 1$
$x_1 = 0.1$	$y_1 = 1.11$
$x_2 = 0.2$	$y_2 = 1.24205$

First-, Second-, and Fourth-Order Approximation

To approximately solve $y' = f(x, y), y(x_0) = y_0$ with step size h

Euler's Method is a First-Order Approximation

It approximates y_{n+1} from y_n using a single-point slope approximation

$$a_n = f(x_n, y_n)h,$$

$$y_{n+1} = y_n + a_n$$

Improved Euler's Method is a Second-Order Approximation

It approximates y_{n+1} from y_n using a two-point slope approximation

$$a_n = f(x_n, y_n)h, \qquad b_n = f(x_{n+1}, y_n + a_n)h$$

$$y_{n+1} = y_n + \frac{1}{2}(a_n + b_n)$$

Fourth-order Runge–Kutta Method

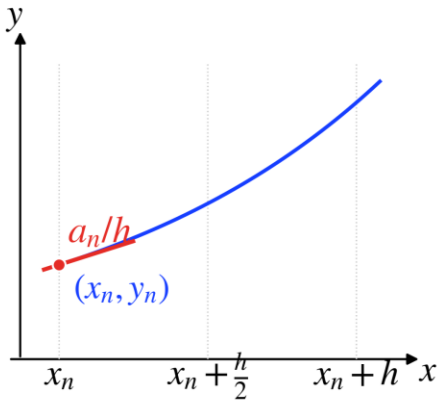
It approximates y_{n+1} from y_n using a four-point slope approximation

$$a_n = f(x_n, y_n)h, \quad b_n = f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}a_n\right)h, \quad c_n = f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}b_n\right)h, \quad d_n = f(x_{n+1}, y_n + c_n)h$$

$$y_{n+1} = y_n + \frac{1}{6}(a_n + 2b_n + 2c_n + d_n)$$

Runge–Kutta Method

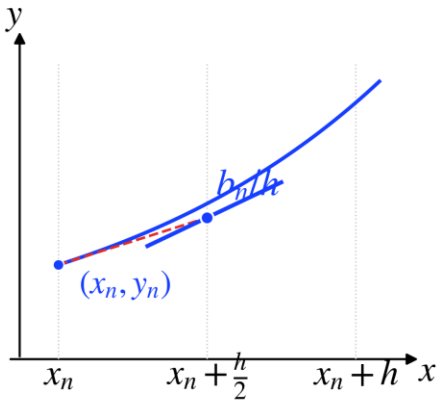
1. Initial slope



Start: use the known point

$$a_n = f(x_n, y_n)h$$

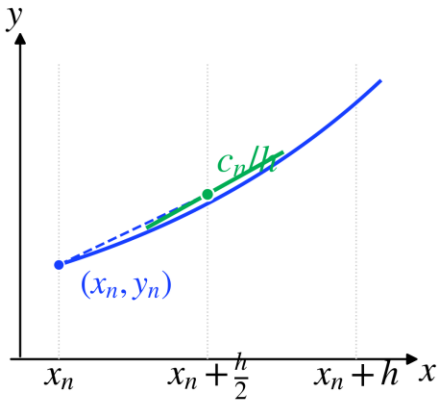
2. First midpoint slope



Predict a midpoint using $a_n/2$

$$b_n = f(x_n + \frac{h}{2},\ y_n + \frac{a_n}{2})h$$

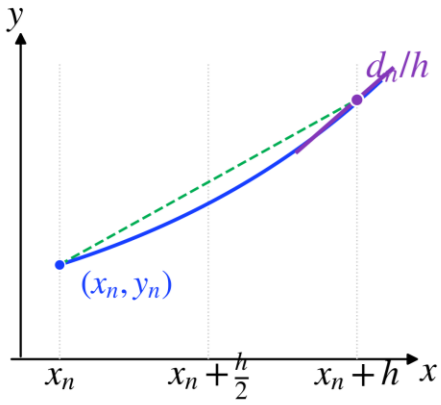
3. Second midpoint slope



Refine the midpoint using $b_n/2$

$$c_n = f(x_n + \frac{h}{2},\ y_n + \frac{b_n}{2})h$$

4. Endpoint slope



Predict an endpoint using c_n

$$d_n = f(x_{n+1},\ y_n + c_n)h$$

$$y_{n+1} = y_n + \frac{1}{6}(a_n + 2b_n + 2c_n + d_n)$$

Fourth-order Runge–Kutta Method

For $y' = f(x, y)$, with $y(x_0) = y_0$, the fourth-order Runge-Kutta method computes approximations $y_n \approx y(x_n)$ at $x_n = x_0 + nh$.

x	a	b	c	d	y
x_0	—	—	—	—	y_0 (known!)
$x_1 = x_0 + h$	$a_0 = f(x_0, y_0)h$	$b_0 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{a_0}{2}\right)h$	$c_0 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{b_0}{2}\right)h$	$d_0 = f(x_1, y_0 + c_0)h$	$y_1 = y_0 + \frac{1}{6}(a_0 + 2b_0 + 2c_0 + d_0)$
$x_2 = x_0 + 2h$	$a_1 = f(x_1, y_1)h$	$b_1 = f\left(x_1 + \frac{h}{2}, y_1 + \frac{a_1}{2}\right)h$	$c_1 = f\left(x_1 + \frac{h}{2}, y_1 + \frac{b_1}{2}\right)h$	$d_1 = f(x_2, y_1 + c_1)h$	$y_2 = y_1 + \frac{1}{6}(a_1 + 2b_1 + 2c_1 + d_1)$
⋮	⋮	⋮	⋮	⋮	⋮
$x_{n+1} = x_0 + (n+1)h$	$a_n = f(x_n, y_n)h$	$b_n = f\left(x_n + \frac{h}{2}, y_n + \frac{a_n}{2}\right)h$	$c_n = f\left(x_n + \frac{h}{2}, y_n + \frac{b_n}{2}\right)h$	$d_n = f(x_{n+1}, y_n + c_n)h$	$y_{n+1} = y_n + \frac{1}{6}(a_n + 2b_n + 2c_n + d_n)$

Fourth-order Runge–Kutta Method

Use fourth-order Runge-Kutta method with step size $h = 0.1$ to approximate $y(0.2)$, given

$$\frac{dy}{dx} = x + y, \qquad y(0) = 1.$$

We have $f(x, y) = x + y$, $x_0 = 0$, and $y_0 = 1$.

First step: $x_1 = 0.1$

$a_0 = f(0, 1) \cdot 0.1 = (0 + 1) \cdot 0.1 = 0.1$

$b_0 = f\left(0.05, 1 + \frac{0.1}{2}\right) \cdot 0.1 = (0.05 + 1.05) \cdot 0.1 = 0.11$

$c_0 = f\left(0.05, 1 + \frac{0.11}{2}\right) \cdot 0.1 = (0.05 + 1.055) \cdot 0.1 = 0.1105$

$d_0 = f(0.1, 1 + 0.1105) \cdot 0.1 = (0.1 + 1.1105) \cdot 0.1 = 0.12105$

$$y_1 = 1 + \frac{0.1 + 2(0.11) + 2(0.1105) + 0.12105}{6} \approx 1.11034167$$

Second step: $x_2 = 0.2$

$a_1 = (0.1 + y_1) \cdot 0.1 \cong 0.12103417$

$b_1 = (0.15 + y_1 + a_1/2) \cdot 0.1 \cong 0.13208588$

$c_1 = (0.15 + y_1 + b_1/2) \cdot 0.1 \cong 0.13263846$

$d_1 = (0.2 + y_1 + c_1) \cdot 0.1 \cong 0.14429801$

$$y_2 \approx 1.11034167 + \frac{0.12103417 + 2(0.13208588) + 2(0.13263846) + 0.14429801}{6} \approx 1.24280514$$

First-, Second-, and Fourth-Order Approximation

Considering the following ODE and initial condition, we can compare the accuracy of different methods.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x + y, \qquad y(0) = 1.$$

Comparison of values:

x	Exact solution $y(x)$	Euler (first order)	Improved Euler (second order)	RK4 (fourth order)
0.0	1.00000000	1.00000000	1.00000000	1.00000000
0.1	1.11034184	1.10000000	1.11000000	1.11034167
0.2	1.24280552	1.22000000	1.24205000	1.24280514

Relative errors:

x	Euler (first order)	Improved Euler (second order)	RK4 (fourth order)
0.0	0%	0%	0%
0.1	0.93141011%	0.03078657%	0.00001526%
0.2	1.83500282%	0.06079119%	0.00003014%

PART II

NUMERICAL METHODS TO SOLVE SECOND-ORDER ODE

Think Like a Mathematician

An engineer, a physicist, and a mathematician are being tested.
They enter a room containing **a table, a kettle, some water, and a stove**.
The kettle is sitting **on the table**.

Question: "How would you boil the water?"

- The engineer says: "Fill the kettle with water, put it on the stove, and heat it."
- The physicist gives essentially the same answer.
- The mathematician also gives the same answer.

Then comes the second problem.
This time, the kettle is **on the floor rather than on the table**.

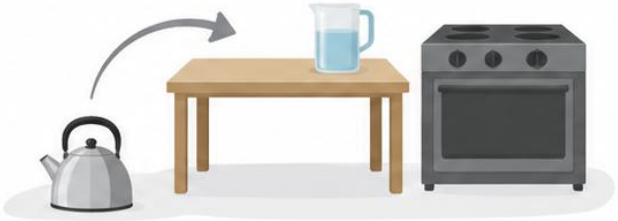
- The engineer picks up the kettle and proceeds as before.
- The physicist does essentially the same thing.
- But the mathematician **picks up the kettle and puts it on the table**.

Interviewer: "What are you doing?"
Mathematician: "I have reduced the problem to one that has already been solved."

Case 1: Kettle on the table



Case 2: Kettle on the floor



In mathematics, we often solve a new problem by transforming it into a problem we already know how to solve.
To solve second-order ODEs, we simply reduce them to a system of first-order ODEs.

Euler's Method for Second-Order ODEs

Use Euler's method with step size $h = 0.1$ to approximate $y(0.2)$, given

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} = x + y, \qquad y(0) = 1, \qquad y'(0) = -1.$$

Our objective is to solve $y'' - y' = x + y$ with the given initial conditions.

Key Idea: We define $y_1 = y$ and $y_2 = y'$.

- From the definitions, we immediately have: $y_1' = y_2$

  Equation 1

- We can rewrite the given second-order ODE as:

$$\underbrace{y''}_{y_2'} - \underbrace{y'}_{y_2} = x + \underbrace{y}_{y_1} \Rightarrow \quad y_2' - y_2 = x + y_1 \quad \Rightarrow \quad y_2' = y_1 + y_2 + x$$

  Equation 2

- We can also rewrite the initial conditions:

$$y(0) = 1 \quad \Rightarrow \quad y_{1,0} = 1 \qquad \text{Initial condition 1}$$

$$y'(0) = -1 \quad \Rightarrow \quad y_{2,0} = -1 \qquad \text{Initial condition 2}$$

Euler's Method for Second-Order ODEs

Use Euler's method with step size $h = 0.1$ to approximate $y(0.2)$, given

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} = x + y, \qquad y(0) = 1, \qquad y'(0) = -1.$$

So by defining $y_1 = y$ and $y_2 = y'$, we can rewrite the problem as:

$$\text{Equations: } \begin{cases} ① \; y_1' = y_2 \\ ② \; y_2' = y_1 + y_2 + x \end{cases} \qquad \text{Initial Conditions: } \begin{cases} y_{1,0} = 1 \\ y_{2,0} = -1 \end{cases}$$

We apply Euler's method :

Equation ①

$$y(0.1) \approx \underbrace{y_{1,1} = y_{1,0} + hy_{1,0}'}_{\text{Euler's method}} = y_{1,0} + hy_{2,0} = 1 + 0.1 \cdot (-1) = 0.9$$

Using the initial conditions

$$y(0.1) \approx 0.9$$

Equation ②

$$y'(0.1) \approx \underbrace{y_{2,1} = y_{2,0} + hy_{2,0}'}_{\text{Euler's method}} = y_{2,0} + h(y_{1,0} + y_{2,0} + x_0) = (-1) + 0.1 \cdot (1 + (-1) + 0) = -1$$

Using the initial conditions

$$y'(0.1) \approx -1$$

Euler's Method for Second-Order ODEs

Use Euler's method with step size $h = 0.1$ to approximate $y(0.2)$, given

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} = x + y, \qquad y(0) = 1, \qquad y'(0) = -1.$$

So by defining $y_1 = y$ and $y_2 = y'$, we can rewrite the problem as:

$$\text{Equations: } \begin{cases} \text{①}\ y_1' = y_2 \\ \text{②}\ y_2' = y_1 + y_2 + x \end{cases} \qquad \text{Initial Conditions: } \begin{cases} y_{1,0} = 1 \\ y_{2,0} = -1 \end{cases}$$

Now we use $y_{1,1} = 0.9$ and $y_{2,1} = -1$ to calculate :

Equation ①

$$y(0.2) \approx \underbrace{y_{1,2} = y_{1,1} + hy_{1,1}'}_{\text{Euler's method}} = y_{1,1} + hy_{2,1} = 0.9 + 0.1 \cdot (-1) = 0.8 \quad \Rightarrow \quad \boxed{y(0.2) \approx 0.8}$$

Using the previous calculations

Final answer!

Equation ②

$$y'(0.2) \approx \underbrace{y_{2,2} = y_{2,1} + hy_{2,1}'}_{\text{Euler's method}} = y_{2,1} + h(y_{1,1} + y_{2,1} + x_1) = (-1) + 0.1 \cdot (0.9 + (-1) + 0.1) = -1$$

Using the previous calculations

Euler's Method for Second-Order ODEs

For $y'' = f(x, y, y')$, with $y(x_0) = y_0$ and $y'(x_0) = y_0'$ Euler's method computes approximations $y_n = y_{1,n} \approx y(x_n)$ and $y_n' = y_{2,n} \approx y'(x_n)$ at $x_n = x_0 + nh$. We have two equations: ① $y_1' = y_2$ ② $y_2' = f(x, y_1, y_2)$

x	$y_{1,n}$	$y_{2,n}$	$y(x)$
x_0	$y_{1,0} = y_0$ (Given initial condition)	$y_{2,0} = y_0'$ (Given initial condition)	$y(x_0) = y_0$ (Given initial condition)
$x_1 = x_0 + h$	$y_{1,1} = y_{1,0} + hy_{1,0}'$ $= y_{1,0} + hy_{2,0}$	$y_{2,1} = y_{2,0} + hy_{2,0}'$ $= y_{2,0} + hf(x_0, y_{1,0}, y_{2,0})$	$y(x_1) \approx y_{1,1}$
$x_2 = x_0 + 2h$	$y_{1,2} = y_{1,1} + hy_{1,1}'$ $= y_{1,1} + hy_{2,1}$	$y_{2,2} = y_{2,1} + hy_{2,1}'$ $= y_{2,1} + hf(x_1, y_{1,1}, y_{2,1})$	$y(x_2) \approx y_{1,2}$
⋮	⋮	⋮	⋮
$x_{n+1} =$ $x_0 + (n+1)h$	$y_{1,n+1} = y_{1,n} + hy_{1,n}'$ $= y_{1,n} + hy_{2,n}$	$y_{2,n+1} = y_{2,n} + hy_{2,n}'$ $= y_{2,n} + hf(x_n, y_{1,n}, y_{2,n})$	$y(x_{n+1}) \approx y_{1,n+1}$

Class Exercise

Use Euler's method with step size $h = 0.2$ to approximate $y(0.6)$, given

$$y'' = -y, \qquad y(0) = 0, \qquad y'(0) = 1.$$

Class Exercise

Use Euler's method with step size $h = 0.2$ to approximate $y(0.6)$, given

$$y'' = -y, \qquad y(0) = 0, \qquad y'(0) = 1.$$

We define $y_1 = y$ and $y_2 = y'$. We have ① $y_1' = y_2$, and ② $y_2' = -y_1$ with initial conditions $y_{1,0} = 0$ and $y_{2,0} = 1$.

x	$y_{1,n}$	$y_{2,n}$	$y(x)$
$x_0 = 0$	$y_{1,0} = y_0 = 0$ (Given initial condition)	$y_{2,0} = y_0' = 1$ (Given initial condition)	$y(x_0) = y_0 = 0$ (Given initial condition)
$x_1 = x_0 + h = 0.2$	$y_{1,1} = y_{1,0} + hy_{1,0}'$ $= y_{1,0} + hy_{2,0} = 0 + 0.2(1)$ $= 0.2$	$y_{2,1} = y_{2,0} + h\,y_{2,0}'$ $= y_{2,0} + h(-y_{1,0})$ $= 1 + 0.2(0) = 1$	$y(x_1) \approx y_{1,1} = 0.2$
$x_2 = x_0 + 2h = 0.4$	$y_{1,2} = y_{1,1} + h\,y_{1,1}'$ $= y_{1,1} + h\,y_{2,1} = 0.2 + 0.2(1)$ $= 0.4$	$y_{2,2} = y_{2,1} + h\,y_{2,1}'$ $= y_{2,1} + h(-y_{1,1})$ $= 1 + 0.2(-0.2) = 0.96$	$y(x_2) \approx y_{1,2} = 0.4$
$x_3 = x_0 + 3h = 0.6$	$y_{1,3} = y_{1,2} + h\,y_{1,2}'$ $= y_{1,2} + h\,y_{2,2}$ $= 0.4 + 0.2(0.96) = 0.592$	$y_{2,3} = y_{2,2} + h\,y_{2,2}'$ $= y_{2,2} + h(-y_{1,2})$ $= 0.96 + 0.2(-0.4) = 0.88$	$\boxed{y(0.6) \approx y_{1,3} = 0.592}$

Improved Euler's Method for Second-Order ODEs

NTU SINGAPORE

For $y'' = f(x, y, y')$, with $y(x_0) = y_0$ and $y'(x_0) = y_0'$, improved Euler's method computes $y_n = y_{1,n} \approx y(x_n)$ and $y_n' = y_{2,n} \approx y'(x_n)$ at $x_n = x_0 + nh$. We have two equations: ① $y_1' = y_2$ ② $y_2' = f(x, y_1, y_2)$

x	a Increment using the slope at the starting point	b Increment using the slope at the predicted endpoint	y_1 and y_2
x_0	—	—	$y_{1,0} = y_0, \quad y_{2,0} = y_0'$ Known initial conditions
$x_1 = x_0 + h$	$a_{1,0} = y_{1,0}'h = y_{2,0}h$ $a_{2,0} = y_{2,0}'h = f(x_0, y_{1,0}, y_{2,0})h$	$b_{1,0} = (y_{2,0} + a_{2,0})h$ $b_{2,0} = f(x_1, y_{1,0} + a_{1,0}, y_{2,0} + a_{2,0})h$	$y_{1,1} = y_{1,0} + \frac{1}{2}(a_{1,0} + b_{1,0})$ $y_{2,1} = y_{2,0} + \frac{1}{2}(a_{2,0} + b_{2,0})$
$x_2 = x_0 + 2h$	$a_{1,1} = y_{1,1}'h = y_{2,1}h$ $a_{2,1} = y_{2,1}'h = f(x_1, y_{1,1}, y_{2,1})h$	$b_{1,1} = (y_{2,1} + a_{2,1})h$ $b_{2,1} = f(x_2, y_{1,1} + a_{1,1}, y_{2,1} + a_{2,1})h$	$y_{1,2} = y_{1,1} + \frac{1}{2}(a_{1,1} + b_{1,1})$ $y_{2,2} = y_{2,1} + \frac{1}{2}(a_{2,1} + b_{2,1})$
⋮	⋮	⋮	⋮
$x_{n+1} = x_0 + (n+1)h$	$a_{1,n} = y_{1,n}'h = y_{2,n}h$ $a_{2,n} = y_{2,n}'h = f(x_n, y_{1,n}, y_{2,n})h$	$b_{1,n} = (y_{2,n} + a_{2,n})h$ $b_{2,n} = f(x_{n+1}, y_{1,n} + a_{1,n}, y_{2,n} + a_{2,n})h$	$y_{1,n+1} = y_{1,n} + \frac{1}{2}(a_{1,n} + b_{1,n})$ $y_{2,n+1} = y_{2,n} + \frac{1}{2}(a_{2,n} + b_{2,n})$

Improved Euler's Method for Second-Order ODEs

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Use improved Euler's method with step size $h = 0.2$ to approximate $y(0.6)$, given $y'' = -y,\ y(0) = 0,\ y'(0) = 1$.

We define $y_1 = y$ and $y_2 = y'$. We have ① $y_1' = y_2$, and ② $y_2' = -y_1$ with initial conditions $y_{1,0} = 0$ and $y_{2,0} = 1$.

x	a Increment using the slope at the starting point	b Increment using the slope at the predicted endpoint	y_1 and y_2
$x_0 = 0$	—	—	$y_{1,0} = 0, \quad y_{2,0} = 1$
$x_1 = 0.2$	$a_{1,0} = y_{2,0}h = (1)(0.2) = 0.2$ $a_{2,0} = -y_{1,0}h = -(0)(0.2) = 0$	$b_{1,0} = (y_{2,0} + a_{2,0})h = (1 + 0)(0.2) = 0.2$ $b_{2,0} = -(y_{1,0} + a_{1,0})h = -(0 + 0.2)(0.2) = -0.04$	$y_{1,1} = y_{1,0} + \frac{1}{2}(a_{1,0} + b_{1,0}) = 0 + \frac{1}{2}(0.2 + 0.2) = 0.2$ $y_{2,1} = y_{2,0} + \frac{1}{2}(a_{2,0} + b_{2,0}) = 1 + \frac{1}{2}(0 - 0.04) = 0.98$
$x_2 = 0.4$	$a_{1,1} = y_{2,1}h = (0.98)(0.2) = 0.196$ $a_{2,1} = -y_{1,1}h = -(0.2)(0.2) = -0.04$	$b_{1,1} = (y_{2,1} + a_{2,1})h = (0.98 - 0.04)(0.2) = 0.188$ $b_{2,1} = -(y_{1,1} + a_{1,1})h = -(0.2 + 0.196)(0.2) = -0.0792$	$y_{1,2} = y_{1,1} + \frac{1}{2}(a_{1,1} + b_{1,1}) = 0.2 + \frac{1}{2}(0.196 + 0.188) = 0.392$ $y_{2,2} = y_{2,1} + \frac{1}{2}(a_{2,1} + b_{2,1}) = 0.98 + \frac{1}{2}(-0.04 - 0.0792) = 0.9204$
$x_3 = 0.6$	$a_{1,2} = y_{2,2}h = (0.9204)(0.2) = 0.18408$ $a_{2,2} = -y_{1,2}h = -(0.392)(0.2) = -0.0784$	$b_{1,2} = (y_{2,2} + a_{2,2})h = (0.9204 - 0.0784)(0.2) = 0.1684$ $b_{2,2} = -(y_{1,2} + a_{1,2})h = -(0.392 + 0.18408)(0.2) = -0.115216$	$y_{1,3} = y_{1,2} + \frac{1}{2}(a_{1,2} + b_{1,2}) = 0.392 + \frac{1}{2}(0.18408 + 0.1684) = 0.56824 \Rightarrow \boxed{y(0.6) \approx 0.56824}$ $y_{2,3} = y_{2,2} + \frac{1}{2}(a_{2,2} + b_{2,2}) = 0.9204 + \frac{1}{2}(-0.0784 - 0.115216) = 0.823592$

First-, Second-, and Fourth-Order Approximation

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To approximately solve $y'' = f(x, y, y'), y(x_0) = y_0, y'(x_0) = y'_0$ with step size h

We define $y_1 = y$ and $y_2 = y'$. We have ① $y'_1 = y_2$, and ② $y'_2 = f(x, y_1, y_2)$ with initial conditions $y_{1,0} = y_0$ and $y_{2,0} = y'_0$.

Euler's Method is a First-Order Approximation

It approximates $y_{1,n+1}$ and $y_{2,n+1}$ using a single-point slope approximation

$$a_{1,n} = y_{2,n}h, \qquad a_{2,n} = f(x_n, y_{1,n}, y_{2,n})h$$

$$y_{1,n+1} = y_{1,n} + a_{1,n}, \qquad y_{2,n+1} = y_{2,n} + a_{2,n}$$

Improved Euler's Method is a Second-Order Approximation

It approximates $y_{1,n+1}$ and $y_{2,n+1}$ using a two-point slope approximation

$$a_{1,n} = y_{2,n}h, \qquad a_{2,n} = f(x_n, y_{1,n}, y_{2,n})h$$

$$b_{1,n} = (y_{2,n} + a_{2,n})h, \qquad b_{2,n} = f(x_{n+1}, y_{1,n} + a_{1,n}\, y_{2,n} + a_{2,n})h$$

$$y_{1,n+1} = y_{1,n} + \frac{1}{2}(a_{1,n} + b_{1,n}), \quad y_{2,n+1} = y_{2,n} + \frac{1}{2}(a_{2,n} + b_{2,n})$$

Fourth-order Runge–Kutta Method

It approximates $y_{1,n+1}$ and $y_{2,n+1}$ using a four-point slope approximation

$$a_{1,n} = y_{2,n}h, \quad a_{2,n} = f(x_n, y_{1,n}, y_{2,n})h, \quad b_{1,n} = \left(y_{2,n} + \frac{1}{2}a_{2,n}\right)h, \quad b_{2,n} = f\left(x_n + \frac{1}{2}h, y_{1,n} + \frac{1}{2}a_{1,n}, y_{2,n} + \frac{1}{2}a_{2,n}\right)h$$

$$c_{1,n} = \left(y_{2,n} + \frac{1}{2}b_{2,n}\right)h, \quad c_{2,n} = f\left(x_n + \frac{1}{2}h, y_{1,n} + \frac{1}{2}b_{1,n}, y_{2,n} + \frac{1}{2}b_{2,n}\right)h, \quad d_{1,n} = (y_{2,n} + c_{2,n})h, \quad d_{2,n} = f(x_{n+1}, y_{1,n} + c_{1,n}, y_{2,n} + c_{2,n})h$$

$$y_{1,n+1} = y_{1,n} + \frac{1}{6}(a_{1,n} + 2b_{1,n} + 2c_{1,n} + d_{1,n}), \quad y_{2,n+1} = y_{2,n} + \frac{1}{6}(a_{2,n} + 2b_{2,n} + 2c_{2,n} + d_{2,n})$$

Fourth-order Runge–Kutta Method

Use 4th–order Runge-Kutta method with step size $h = 0.2$ to approximate $y(0.6)$, given $y'' = -y,\ y(0) = 0,\ y'(0) = 1$.

We define $y_1 = y$ and $y_2 = y'$. We have ① $y_1' = y_2$, and ② $y_2' = -y_1$ with initial conditions $y_{1,0} = 0$ and $y_{2,0} = 1$.

$$a_{1,n} = y_{2,n}h, \qquad a_{2,n} = -y_{1,n}h,$$
$$b_{1,n} = \left(y_{2,n} + \frac{1}{2}a_{2,n}\right)h, \qquad b_{2,n} = -\left(y_{1,n} + \frac{1}{2}a_{1,n}\right)h,$$
$$c_{1,n} = \left(y_{2,n} + \frac{1}{2}b_{2,n}\right)h, \qquad c_{2,n} = -\left(y_{1,n} + \frac{1}{2}b_{1,n}\right)h,$$
$$d_{1,n} = \left(y_{2,n} + c_{2,n}\right)h, \qquad d_{2,n} = -\left(y_{1,n} + c_{1,n}\right)h,$$

$$y_{1,n+1} = y_{1,n} + \frac{1}{6}\left(a_{1,n} + 2b_{1,n} + 2c_{1,n} + d_{1,n}\right), \quad y_{2,n+1} = y_{2,n} + \frac{1}{6}\left(a_{2,n} + 2b_{2,n} + 2c_{2,n} + d_{2,n}\right)$$

At $x_1 = 0.2$

$$a_{1,0} = 1(0.2) = 0.2, \qquad a_{2,0} = -(0)(0.2) = 0$$
$$b_{1,0} = (1 + 0)(0.2) = 0.2, \qquad b_{2,0} = -(0 + 0.1)(0.2) = -0.02$$
$$c_{1,0} = (1 - 0.01)(0.2) = 0.198, \qquad c_{2,0} = -(0 + 0.1)(0.2) = -0.02$$
$$d_{1,0} = (1 - 0.02)(0.2) = 0.196, \qquad d_{2,0} = -(0 + 0.198)(0.2) = -0.0396$$

Therefore:

$$y(0.2) \approx y_{1,1} = 0 + \frac{1}{6}(0.2 + 2(0.2) + 2(0.198) + 0.196) = 0.1986667$$
$$y'(0.2) \approx y_{2,1} = 1 + \frac{1}{6}(0 + 2(-0.02) + 2(-0.02) - 0.0396) = 0.9800667$$

Fourth-order Runge–Kutta Method

Use 4th–order Runge-Kutta method with step size $h = 0.2$ to approximate $y(0.6)$, given $y'' = -y,\ y(0) = 0,\ y'(0) = 1$.

We define $y_1 = y$ and $y_2 = y'$. We have ① $y_1' = y_2$, and ② $y_2' = -y_1$ with initial conditions $y_{1,0} = 0$ and $y_{2,0} = 1$.

$$a_{1,n} = y_{2,n}h, \qquad a_{2,n} = -y_{1,n}h,$$
$$b_{1,n} = \left(y_{2,n} + \frac{1}{2}a_{2,n}\right)h, \qquad b_{2,n} = -\left(y_{1,n} + \frac{1}{2}a_{1,n}\right)h,$$
$$c_{1,n} = \left(y_{2,n} + \frac{1}{2}b_{2,n}\right)h, \qquad c_{2,n} = -\left(y_{1,n} + \frac{1}{2}b_{1,n}\right)h,$$
$$d_{1,n} = (y_{2,n} + c_{2,n})h, \qquad d_{2,n} = -(y_{1,n} + c_{1,n})h,$$

$$y_{1,n+1} = y_{1,n} + \frac{1}{6}(a_{1,n} + 2b_{1,n} + 2c_{1,n} + d_{1,n}), \quad y_{2,n+1} = y_{2,n} + \frac{1}{6}(a_{2,n} + 2b_{2,n} + 2c_{2,n} + d_{2,n})$$

At $x_2 = 0.4$

$a_{1,1} = (0.9800667)(0.2) = 0.1960133,$
$b_{1,1} = (0.9800667 - 0.0198667)(0.2) = 0.1920400,$
$c_{1,1} = (0.9800667 - 0.0296673)(0.2) = 0.1900799,$
$d_{1,1} = (0.9800667 - 0.0589373)(0.2) = 0.1842259,$

$a_{2,1} = -(0.1986667)(0.2) = -0.0397333$
$b_{2,1} = -(0.1986667 + 0.0980067)(0.2) = -0.0593347$
$c_{2,1} = -(0.1986667 + 0.0960200)(0.2) = -0.0589373$
$d_{2,1} = -(0.1986667 + 0.1900799)(0.2) = -0.0777493$

Therefore:

$$y(0.4) \approx y_{1,2} = 0.1986667 + \frac{1}{6}(0.1960133 + 2(0.1920400) + 2(0.1900799) + 0.1842259) = 0.3894132$$
$$y'(0.4) \approx y_{2,2} = 0.9800667 + \frac{1}{6}(-0.0397333 + 2(-0.0593347) + 2(-0.0589373) - 0.0777493) = 0.9210622$$

Fourth-order Runge–Kutta Method

Use 4th–order Runge-Kutta method with step size $h = 0.2$ to approximate $y(0.6)$, given $y'' = -y,\ y(0) = 0,\ y'(0) = 1$.

We define $y_1 = y$ and $y_2 = y'$. We have ① $y_1' = y_2$, and ② $y_2' = -y_1$ with initial conditions $y_{1,0} = 0$ and $y_{2,0} = 1$.

$$a_{1,n} = y_{2,n}h, \qquad a_{2,n} = -y_{1,n}h,$$
$$b_{1,n} = \left(y_{2,n} + \frac{1}{2}a_{2,n}\right)h, \qquad b_{2,n} = -\left(y_{1,n} + \frac{1}{2}a_{1,n}\right)h,$$
$$c_{1,n} = \left(y_{2,n} + \frac{1}{2}b_{2,n}\right)h, \qquad c_{2,n} = -\left(y_{1,n} + \frac{1}{2}b_{1,n}\right)h,$$
$$d_{1,n} = (y_{2,n} + c_{2,n})h, \qquad d_{2,n} = -(y_{1,n} + c_{1,n})h,$$

$$y_{1,n+1} = y_{1,n} + \frac{1}{6}(a_{1,n} + 2b_{1,n} + 2c_{1,n} + d_{1,n}), \quad y_{2,n+1} = y_{2,n} + \frac{1}{6}(a_{2,n} + 2b_{2,n} + 2c_{2,n} + d_{2,n})$$

At $x_3 = 0.6$

$a_{1,2} = (0.9210622)(0.2) = 0.1842124,$
$b_{1,2} = (0.9210622 - 0.0389413)(0.2) = 0.1764242,$
$c_{1,2} = (0.9210622 - 0.0481519)(0.2) = 0.1745821,$
$d_{1,2} = (0.9210622 - 0.0955251)(0.2) = 0.1651074,$

$a_{2,2} = -(0.3894132)(0.2) = -0.0778826$
$b_{2,2} = -(0.3894132 + 0.0921062)(0.2) = -0.0963039$
$c_{2,2} = -(0.3894132 + 0.0882121)(0.2) = -0.0955251$
$d_{2,2} = -(0.3894132 + 0.1745821)(0.2) = -0.1127991$

Therefore:

$$\boldsymbol{y(0.6) \approx y_{1,3} = 0.3894132 + \frac{1}{6}(0.1842124 + 2(0.1764242) + 2(0.1745821) + 0.1651074) = \boxed{0.5646353}}$$

$$y'(0.6) \approx y_{2,3} = 0.9210622 + \frac{1}{6}(-0.0778826 + 2(-0.0963039) + 2(-0.0955251) - 0.1127991) = 0.8253389$$

First-, Second-, and Fourth-Order Approximation

Considering $y'' = -y,\ y(0) = 0,\ y'(0) = 1$, we can compare the accuracy of different methods.

Comparison of values:

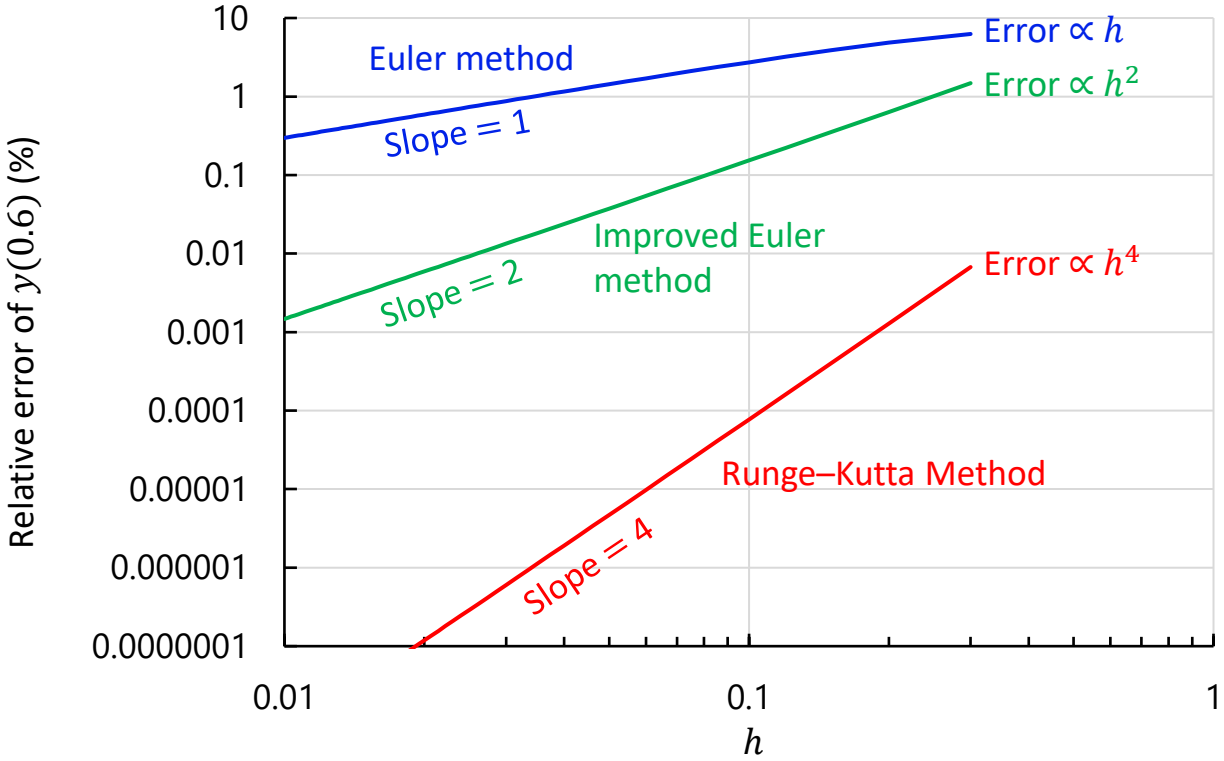
x	Exact solution $y(x)$	Euler (first order)	Improved Euler (second order)	Runge–Kutta (fourth order)
0.0	0.00000000	0.00000000	0.00000000	0.00000000
0.2	0.19866933	0.20000000	0.20000000	0.19866670
0.4	0.38941834	0.40000000	0.39200000	0.38941320
0.6	0.56464247	0.59200000	0.56824000	0.56463530

Relative errors:

x	Euler (first order)	Improved Euler (second order)	Runge–Kutta (fourth order)
0.2	0.66979095%	0.66979095%	0.00132421%
0.4	2.71729822%	0.66295226%	0.00132051%
0.6	4.84510604%	0.63713354%	0.00127043%

First-, Second-, and Fourth-Order Approximation

Using different numerical method with step size h to approximate $y(0.6)$, given $y'' = -y,\ y(0) = 0,\ y'(0) = 1.$



Summary: First-Order ODEs

To approximately solve $y' = f(x, y), y(x_0) = y_0$ with step size h

Euler's Method is a First-Order Approximation

It approximates y_{n+1} from y_n using a single-point slope approximation

$$a_n = f(x_n, y_n)h,$$

$$y_{n+1} = y_n + a_n$$

Improved Euler's Method is a Second-Order Approximation

It approximates y_{n+1} from y_n using a two-point slope approximation

$$a_n = f(x_n, y_n)h, \qquad b_n = f(x_{n+1}, y_n + a_n)h$$

$$y_{n+1} = y_n + \frac{1}{2}(a_n + b_n)$$

Fourth-order Runge–Kutta Method

It approximates y_{n+1} from y_n using a four-point slope approximation

$$a_n = f(x_n, y_n)h, \quad b_n = f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}a_n\right)h, \quad c_n = f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}b_n\right)h, \quad d_n = f(x_{n+1}, y_n + c_n)h$$

$$y_{n+1} = y_n + \frac{1}{6}(a_n + 2b_n + 2c_n + d_n)$$

Summary: Second-Order ODEs

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To approximately solve $y'' = f(x, y, y'), y(x_0) = y_0, y'(x_0) = y'_0$ with step size h

We define $y_1 = y$ and $y_2 = y'$. We have ① $y'_1 = y_2$, and ② $y'_2 = f(x, y_1, y_2)$ with initial conditions $y_{1,0} = y_0$ and $y_{2,0} = y'_0$.

Euler's Method is a First-Order Approximation

It approximates $y_{1,n+1}$ and $y_{2,n+1}$ using a single-point slope approximation

$$a_{1,n} = y_{2,n}h, \qquad a_{2,n} = f(x_n, y_{1,n}, y_{2,n})h$$

$$y_{1,n+1} = y_{1,n} + a_{1,n}, \qquad y_{2,n+1} = y_{2,n} + a_{2,n}$$

Improved Euler's Method is a Second-Order Approximation

It approximates $y_{1,n+1}$ and $y_{2,n+1}$ using a two-point slope approximation

$$a_{1,n} = y_{2,n}h, \qquad a_{2,n} = f(x_n, y_{1,n}, y_{2,n})h$$

$$b_{1,n} = (y_{2,n} + a_{2,n})h, \qquad b_{2,n} = f(x_{n+1}, y_{1,n} + a_{1,n}\, y_{2,n} + a_{2,n})h$$

$$y_{1,n+1} = y_{1,n} + \frac{1}{2}(a_{1,n} + b_{1,n}), \quad y_{2,n+1} = y_{2,n} + \frac{1}{2}(a_{2,n} + b_{2,n})$$

Fourth-order Runge–Kutta Method

It approximates $y_{1,n+1}$ and $y_{2,n+1}$ using a four-point slope approximation

$$a_{1,n} = y_{2,n}h, \quad a_{2,n} = f(x_n, y_{1,n}, y_{2,n})h, \quad b_{1,n} = \left(y_{2,n} + \frac{1}{2}a_{2,n}\right)h, \quad b_{2,n} = f\left(x_n + \frac{1}{2}h, y_{1,n} + \frac{1}{2}a_{1,n}, y_{2,n} + \frac{1}{2}a_{2,n}\right)h$$

$$c_{1,n} = \left(y_{2,n} + \frac{1}{2}b_{2,n}\right)h, \quad c_{2,n} = f\left(x_n + \frac{1}{2}h, y_{1,n} + \frac{1}{2}b_{1,n}, y_{2,n} + \frac{1}{2}b_{2,n}\right)h, \quad d_{1,n} = (y_{2,n} + c_{2,n})h, \quad d_{2,n} = f(x_{n+1}, y_{1,n} + c_{1,n}, y_{2,n} + c_{2,n})h$$

$$y_{1,n+1} = y_{1,n} + \frac{1}{6}(a_{1,n} + 2b_{1,n} + 2c_{1,n} + d_{1,n}), \quad y_{2,n+1} = y_{2,n} + \frac{1}{6}(a_{2,n} + 2b_{2,n} + 2c_{2,n} + d_{2,n})$$

Tutorial Questions

1. Consider the initial value problem

$$y' = \frac{1}{2}\left(\frac{y}{x} - \frac{x}{y}\right), \qquad y(2) = 2.$$

Use the following methods with step size $h = 0.2$ to find $y(2.6)$.

(a) Euler method

(b) Improved Euler method

(c) Runge-Kutta method

(Exact solution is $y = \sqrt{4x - x^2}$)

2. Consider the initial value problem

$$y'' + 2(y')^2 + 10y = 0$$

$$y(0) = 0, \qquad y'(0) = 1.$$

Use Euler method with step size $h = 0.1$ to approximate $y(0.2)$ and $y'(0.2)$.

3. Consider the initial value problem

$$y''(x) - xy'(x) + y(x) = x^2$$

$$y(2) = 3, \qquad y'(2) = 0$$

where $y''(x) = \frac{d^2y}{dx^2}$ and $y'(x) = \frac{dy}{dx}$. Use improved Euler method (Heun's method) with step size $h = 0.1$ to find approximate values of $y(2.3)$ and $y'(2.3)$.

4. Apply Runge-Kutta method with step size $h = 0.04$ to the initial value problem (here $y'' = \frac{d^2y}{dt^2}$)

$$y'' = 5 + 2ty$$

$$y(0) = 2, \qquad y'(0) = 0.$$

Find approximate values of $y(0.12)$ and $y'(0.12)$.